

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{12} g_2^2 (9 g_1^2 + 13 g_2^2) + \frac{2}{9} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,0}[m_2] + \frac{1}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,-1}[m_2] - \right. \\ & \frac{16}{45} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{5,-2}[m_2] + \frac{1}{4} (-2 g_2^2 \overline{y_e^{PR}} y_e^{PR} + \sum p g_2^4) LF_{2,0}[m_1^P] + \\ & \frac{1}{36} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (-18 \overline{y_e^{PR}} y_e^{PR} + 13 \sum p g_2^2) LF_{3,0}[m_1^P] + \\ & \frac{1}{24} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (12 \overline{y_e^{PR}} y_e^{PR} - 11 \sum p g_2^2) LF_{4,-1}[m_1^P] + \\ & \frac{4}{45} \sum p g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{5,-2}[m_1^P] + \frac{3}{4} g_2^2 (-2 \overline{y_d^{PR}} y_d^{PR} - 2 \overline{y_u^{PR}} y_u^{PR} + \sum p g_2^2) LF_{2,0}[m_q^P] + \\ & \frac{1}{12} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (-18 (\overline{y_d^{PR}} y_d^{PR} + \overline{y_u^{PR}} y_u^{PR}) + 13 \sum p g_2^2) LF_{3,0}[m_q^P] + \\ & \frac{1}{8} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (12 \overline{y_d^{PR}} y_d^{PR} + 12 \overline{y_u^{PR}} y_u^{PR} - 11 \sum p g_2^2) LF_{4,-1}[m_q^P] + \\ & \frac{4}{15} \sum p g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{5,-2}[m_q^P] + \frac{1}{9} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,0}[\tilde{\mu}] + \\ & \frac{1}{6} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,-1}[\tilde{\mu}] - \frac{8}{45} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{5,-2}[\tilde{\mu}] + g_1^4 \tilde{\mu}^2 LF_{2,2,-1}[m_1, \tilde{\mu}] + \\ & \frac{1}{2} m_1 \tilde{\mu} g_1^4 (\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2}) LF_{2,2,0}[m_1, \tilde{\mu}] - \frac{4}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{2,1,0}[m_2, \tilde{\mu}] - \\ & 2 g_2^4 LF_{2,2,-2}[m_2, \tilde{\mu}] + \frac{1}{3} g_2^4 (2 (C_{\phi 1^2} + C_{\phi 2^2} + 3 m_2^2) + 9 \tilde{\mu}^2) LF_{2,2,-1}[m_2, \tilde{\mu}] + \\ & \frac{1}{6} m_2 g_2^4 (-6 m_2 (C_{\phi 1^2} + C_{\phi 2^2}) + 13 \tilde{\mu} (\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2})) LF_{2,2,0}[m_2, \tilde{\mu}] + \\ & \frac{2}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,1,-1}[m_2, \tilde{\mu}] - \frac{10}{3} g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,2,-2}[m_2, \tilde{\mu}] + \\ & \frac{1}{3} m_2 g_2^4 (9 m_2 (C_{\phi 1^2} + C_{\phi 2^2}) - \tilde{\mu} (\overline{C_{\phi 1\phi 2}} + C_{\phi 1\phi 2})) LF_{3,2,-1}[m_2, \tilde{\mu}] + \\ & 2 g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,3,-3}[m_2, \tilde{\mu}] - 2 g_2^4 m_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,3,-2}[m_2, \tilde{\mu}] + \\ & 2 g_2^4 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,2,-3}[m_2, \tilde{\mu}] - 2 g_2^4 m_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) LF_{4,2,-2}[m_2, \tilde{\mu}] + \\ & 3 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} LF_{1,1,0}[m_d^r, m_u^t] - g_2^2 \overline{a_e^{PR}} a_e^{PR} LF_{2,1,0}[m_l^P, m_e^r] + \\ & \frac{1}{2} g_2^2 (\overline{a_e^{PR}} a_e^{PR} + \tilde{\mu}^2 \overline{y_e^{PR}} y_e^{PR}) LF_{3,1,-1}[m_l^P, m_e^r] - \\ & g_2^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_e^{PR}} a_e^{PR} + \overline{a_e^{PR}} (2 C_{\phi 2^2} a_e^{PR} + C_{\phi 1\phi 2} \tilde{\mu} y_e^{PR})) LF_{3,1,0}[m_l^P, m_e^r] + \\ & \frac{1}{4} g_2^2 (\overline{a_e^{PR}} (18 C_{\phi 2^2} a_e^{PR} + 13 C_{\phi 1\phi 2} \tilde{\mu} y_e^{PR}) + \tilde{\mu} \overline{y_e^{PR}} (13 \overline{C_{\phi 1\phi 2}} a_e^{PR} + 2 \tilde{\mu} y_e^{PR} (4 C_{\phi 1^2} + C_{\phi 2^2}))) \\ & LF_{4,1,-1}[m_l^P, m_e^r] - \frac{1}{3} g_2^2 (\overline{a_e^{PR}} (a_e^{PR} (C_{\phi 1^2} + 7 C_{\phi 2^2}) + 7 C_{\phi 1\phi 2} \tilde{\mu} y_e^{PR}) + \\ & \tilde{\mu} \overline{y_e^{PR}} (7 \overline{C_{\phi 1\phi 2}} a_e^{PR} + \tilde{\mu} y_e^{PR} (7 C_{\phi 1^2} + C_{\phi 2^2}))) LF_{5,1,-2}[m_l^P, m_e^r] - \\ & 3 g_2^2 \overline{a_d^{PR}} a_d^{PR} LF_{2,1,0}[m_q^P, m_d^r] + \frac{3}{2} g_2^2 (\overline{a_d^{PR}} a_d^{PR} + \tilde{\mu}^2 \overline{y_d^{PR}} y_d^{PR}) LF_{3,1,-1}[m_q^P, m_d^r] - \\ & 3 g_2^2 (\overline{C_{\phi 1\phi 2}} \tilde{\mu} \overline{y_d^{PR}} a_d^{PR} + \overline{a_d^{PR}} (2 C_{\phi 2^2} a_d^{PR} + C_{\phi 1\phi 2} \tilde{\mu} y_d^{PR})) LF_{3,1,0}[m_q^P, m_d^r] + \\ & \frac{3}{4} g_2^2 (\overline{a_d^{PR}} (18 C_{\phi 2^2} a_d^{PR} + 13 C_{\phi 1\phi 2} \tilde{\mu} y_d^{PR}) + \tilde{\mu} \overline{y_d^{PR}} (13 \overline{C_{\phi 1\phi 2}} a_d^{PR} + 2 \tilde{\mu} y_d^{PR} (4 C_{\phi 1^2} + C_{\phi 2^2}))) \\ & LF_{4,1,-1}[m_q^P, m_d^r] - \\ & g_2^2 (\overline{a_d^{PR}} (a_d^{PR} (C_{\phi 1^2} + 7 C_{\phi 2^2}) + 7 C_{\phi 1\phi 2} \tilde{\mu} y_d^{PR}) + \tilde{\mu} \overline{y_d^{PR}} (7 \overline{C_{\phi 1\phi 2}} a_d^{PR} + \tilde{\mu} y_d^{PR} (7 C_{\phi 1^2} + C_{\phi 2^2}))) \\ & LF_{5,1,-2}[m_q^P, m_d^r] + 3 \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} LF_{1,1,0}[m_q^P, m_q^s] + \\ & \frac{3}{2} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (C_{\phi 1^2} + C_{\phi 2^2}) LF_{2,1,0}[m_q^P, m_q^s] - \frac{3}{2} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (C_{\phi 1^2} + C_{\phi 2^2}) \\ & LF_{3,1,-1}[m_q^P, m_q^s] + \frac{3}{2} g_2^2 (-\overline{a_u^{PR}} a_u^{PR} + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR}) LF_{2,1,0}[m_q^P, m_u^r] + \\ & \frac{1}{2} g_2^2 (\overline{a_u^{PR}} (a_u^{PR} (-2 C_{\phi 1^2} + C_{\phi 2^2}) + C_{\phi 1\phi 2} \tilde{\mu} y_u^{PR}) + \tilde{\mu} \overline{y_u^{PR}} (\overline{C_{\phi 1\phi 2}} a_u^{PR} + \tilde{\mu} y_u^{PR} (C_{\phi 1^2} + 4 C_{\phi 2^2}))) \\ & LF_{2,2,0}[m_q^P, m_u^r] - \frac{3}{2} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (\overline{a_u^{PR}} a_u^{PR} - \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR}) LF_{3,1,0}[m_q^P, m_u^r] + \\ & \frac{1}{4} g_2^2 (\overline{a_u^{PR}} (2 a_u^{PR} (C_{\phi 1^2} + C_{\phi 2^2}) - C_{\phi 1\phi 2} \tilde{\mu} y_u^{PR}) - \tilde{\mu} \overline{y_u^{PR}} (\overline{C_{\phi 1\phi 2}} a_u^{PR} + 4 \tilde{\mu} y_u^{PR} (C_{\phi 1^2} + C_{\phi 2^2}))) \\ & LF_{3,2,-1}[m_q^P, m_u^r] + \frac{3}{2} g_2^2 (C_{\phi 1^2} + C_{\phi 2^2}) (\overline{a_u^{PR}} a_u^{PR} - \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR}) LF_{4,1,-1}[m_q^P, m_u^r] + \\ & \frac{3}{2} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (C_{\phi 1^2} + C_{\phi 2^2}) LF_{2,1,0}[m_q^s, m_q^p] - \\ & \frac{3}{2} \overline{y_d^{PR}} y_d^{SR} \overline{y_u^{ST}} y_u^{PT} (C_{\phi 1^2} + C_{\phi 2^2}) LF_{3,1,-1}[m_q^s, m_q^p] - \frac{3}{2} g_2^2 (\overline{a_u^{PR}} a_u^{PR} + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR}) \\ & LF_{2,1,0}[m_u^r, m_q^p] + \frac{3}{2} g_2^2 (\overline{a_u^{PR}} a_u^{PR} + \tilde{\mu}^2 \overline{y_u^{PR}} y_u^{PR}) LF_{3,1,-1}[m_u^r, m_q^p] - \\ & \frac{1}{2} g_2^2 (\overline{a_u^{PR}} (a_u^{PR} (7 C_{\phi 1^2} + C_{\phi 2^2}) + 7 C_{\phi 1\phi 2} \tilde{\mu} y_u^{PR}) + \tilde{\mu} \overline{y_u^{PR}} (7 \overline{C_{\phi 1\phi 2}} a_u^{PR} + \tilde{\mu} y_u^{PR} (C_{\phi 1^2} + 7 C_{\phi 2^2}))) \\ & LF_{3,1,0}[m_u^r, m_q$$